

|              |   |           |   |       |                        |       |        |
|--------------|---|-----------|---|-------|------------------------|-------|--------|
| CUSTOMER     | x |           |   |       |                        |       |        |
| PROJECT      | x |           |   |       |                        |       |        |
| MANUFACTURER | x |           |   |       |                        |       |        |
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|                           | Unit       | Reference    | Clause  | Symbol | Formula            | TREFOIL<br>GND | TREFOIL<br>DUCT | TREFOIL<br>AIR |
|---------------------------|------------|--------------|---------|--------|--------------------|----------------|-----------------|----------------|
| DESIGN DATA               |            |              |         |        |                    |                |                 |                |
| Bonding Type              |            | Design data  |         |        |                    | SOLID          | SOLID           | SOLID          |
| Cond Mat                  |            | Design data  |         |        |                    | Cu             | Cu              | Cu             |
| Cond size                 | mm2        | Design data  |         |        |                    | 185            | 185             | 185            |
| Cond design               |            | Design data  |         | cc     |                    | stranded       | stranded        | stranded       |
| Cond Dn                   | mm         | Design data  |         | Dc     |                    | 15.9           | 15.9            | 15.9           |
| Cond Sc Dn                | mm         | Design data  |         | Ds     |                    | 17.6           | 17.6            | 17.6           |
| Ins Mat                   |            | Design data  |         |        |                    | XLPE           | XLPE            | XLPE           |
| Ins Dn                    | mm         | Design data  |         | Di     |                    | 33.9           | 33.9            | 33.9           |
| Ins Sc Dn                 | mm         | Design data  |         | dx     |                    | 35.6           | 35.6            | 35.6           |
| Ins Tn                    | mm         | Design data  |         | Ti     |                    | 8.15           | 8.15            | 8.15           |
| Cable Dn                  | mm         | Design data  |         | De     |                    | 43.4           | 43.4            | 43.4           |
| Max cond temp             | C          | Design Data  |         | Tmax   |                    | 90             | 90              | 90             |
| Rated voltage             | kV         | IEC60502-2   | Table 1 | U      |                    | 33             | 33              | 33             |
| Earth voltage             | kV         | IEC60502-2   | Table 1 | Uo     |                    | 19.05          | 19.05           | 19.05          |
| PARAMETER SETTING         |            |              |         |        |                    |                |                 |                |
| Soil TR                   | K.m/W      | Input        |         | p      |                    | 1.20           | 1.20            | --             |
| Frequency                 | Hz         | Input        |         | f      |                    | 50             | 50              | 50             |
| Ambient temp.             | C          | Input        |         | Ta     |                    | 30             | 30              | 40             |
| Depth                     | mm         | Input        |         | L      |                    | 1000           | 1000            | --             |
| Duct Type                 |            | Input        |         |        |                    | --             | HDPE            | --             |
| External diam of duct     | mm         | Input        |         | dc     |                    | --             | 110.0           | --             |
| Thickness of duct         | mm         | Input        |         | Td     |                    | --             | 5.0             | --             |
| Temp rise, amb to max.    | C          | IEC60287-1-1 | 1.4.1.1 | dT     | Tmax - Ta          | 60             | 60              | 50             |
| Core temperature, Tc      | C          | Iteration    |         | Tc     |                    | 90.00          | 90.00           | 90.00          |
| Ins Relative permittivity |            | IEC60287-1-1 | Table 3 | er     |                    | 2.500          | 2.500           | 2.500          |
| Loss factor               |            | IEC60287-1-1 | Table 3 | tand   |                    | 0.0010         | 0.0010          | 0.0010         |
| Dielectric loss           | W/m/p<br>h | EC60287-1-1  | 2.2     | Wd     | wCUo2tand          | 0.0242         | 0.0242          | 0.0242         |
| DC RESISTANCE             |            |              |         |        |                    |                |                 |                |
| DC resistant at 20C       | O/m        | IEC60228     | Class 2 | Ro     |                    | 9.910E-05      | 9.910E-05       | 9.910E-05      |
| DC resistant at ambient   | O/m        | IEC60287-1-1 | 2.1.1   | Ro amb | Ro(1+a20(Ta-20))   | 1.030E-04      | 1.030E-04       | 1.030E-04      |
| DC resistant at 90C       | O/m        | IEC60287-1-1 | 2.1.1   | Ro max | Ro(1+a20(Tmax-20)) | 1.264E-04      | 1.264E-04       | 1.264E-04      |
| AC RESISTANCE             |            |              |         |        |                    |                |                 |                |
| Strand constant           |            | IEC60287-1-1 | Table 2 | ks     |                    | 1.000          | 1.000           | 1.000          |

|                 | Unit | Reference    | Clause  | Symbol | Formula                        | TREFOIL<br>GND | TREFOIL<br>DUCT | TREFOIL<br>AIR |
|-----------------|------|--------------|---------|--------|--------------------------------|----------------|-----------------|----------------|
| Strand constant |      | IEC60287-1-1 | Table 2 | kp     |                                | 0.800          | 0.800           | 0.800          |
| Bessel function |      | IEC60287-1-1 | 2.1.2   | Xs4    | $[k_s(8\pi i f/R_o)10^{-7}]^2$ | 0.989          | 0.989           | 0.989          |
| Skin effect     |      | IEC60287-1-1 | 2.1.2   | Ys     | $Xs4/(192+0.8 Xs4)$            | 0.005          | 0.005           | 0.005          |

|              |           |   |       |                        |       |        |  |  |
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|                     | Unit | Reference    | Clause | Symbol | Formula                                      | TREFOIL<br>GND | TREFOIL<br>DUCT | TREFOIL<br>AIR |
|---------------------|------|--------------|--------|--------|----------------------------------------------|----------------|-----------------|----------------|
| Proximity function  |      | IEC60287-1-1 | 2.1.4  | Xp4    | $[kp(8pif/Ro)10^{-7}]^2$                     | 0.633          | 0.633           | 0.633          |
| Proximity function  |      | IEC60287-1-1 | 2.1.4  | Yp"    | $Xp4/(192+0.8 Xp4)$                          | 0.003          | 0.003           | 0.003          |
| Proximity effect    |      | IEC60287-1-1 | 2.1.4  | Yp     | $Yp"(Dc/s)^2[0.312(Dc/s)^2+1.18/(Yp"+0.27)]$ | 0.002          | 0.000           | 0.002          |
| AC resistant at 90C | O/m  | IEC60287-1-1 | 2.1    | R90    | $Ro \max(1+Ys+Yp)$                           | 1.273E-04      | 1.270E-04       | 1.273E-04      |

SHEATH LOSS FACTOR

|                                 |     |              |         |     |                                          |           |           |           |
|---------------------------------|-----|--------------|---------|-----|------------------------------------------|-----------|-----------|-----------|
| Equivalent mean diameter        | mm  | Design data  |         | ds  |                                          | 36.0      | 36.0      | 36.0      |
| Equivalent resistant at 20C     | O/m | IEC60287-1-1 |         | R20 | p/A                                      | 3.448E-04 | 3.448E-04 | 3.448E-04 |
| Equivalent resistant at Tc      | O/m | IEC60287-1-1 |         | Rs  | $Rs0(1+a20(Tc-20))$                      | 4.275E-04 | 4.292E-04 | 4.212E-04 |
| Sheath Reactance                | O/m | IEC60287-1-1 | 2.3.3   | X   | $2wln(2s/d)x10^{-7}$                     | 5.472E-05 | 1.137E-04 | 5.472E-05 |
| Circulating current loss factor |     | IEC60287-1-1 | 2.3.1   | l1' | $Rs/R/(1+(Rs/X)^2)$                      | 0.0541    | 0.2216    | 0.0549    |
| Eddy current loss factor        |     | IEC60287-1-1 | 2.3.6.1 | l1" | $(Rs/R)gslo(1+D1+D2)+(b1ts)^4/(12x1010)$ | 0.0000    | 0.0000    | 0.0000    |
| Sheath loss factor              |     | IEC60287-1-1 | 2.3     | l1  | $l1'+l1''$                               | 0.0541    | 0.2216    | 0.0549    |

THERMAL RESISTANCE

|                                 |       |              |       |       |                            |       |        |       |
|---------------------------------|-------|--------------|-------|-------|----------------------------|-------|--------|-------|
| T1 factor                       |       | IEC60287-2-1 |       |       |                            | 1.000 | 1.000  | 1.000 |
| Thermal Resistant Insulation    | K.m/W | IEC60287-2-1 |       | T1    | $(p/2p)LN[1+(2t/dc)]$      | 0.451 | 0.451  | 0.451 |
| T3 factor                       |       | IEC60287-2-1 |       |       |                            | 1.600 | 1.000  | 1.000 |
| Thermal Resistant Outercovering | K.m/W | IEC60287-2-1 |       | T3    | $(p/2p)LN[1+(2t/D)]$       | 0.208 | 0.130  | 0.130 |
| T4' air gap                     | K.m/W | IEC60287-2-1 | 2.2.7 | T4'   | $U/(1+0.1(V+YTm)De)$       | --    | 0.5277 | --    |
| T4" duct wall                   | K.m/W | IEC60287-2-1 | 2.2.7 | T4"   | $(pT/2p)LN(Do/Dd)$         | --    | 0.0531 | --    |
| T4''' soil                      | K.m/W | IEC60287-2-1 | 2.2.7 | T4''' | $Tself+2xTmutual$          | --    | 1.7946 | --    |
| Thermal Resistant External      | K.m/W | IEC60287-2-1 | 2.2   | T4    | $(pT/2p)LN(u+SQRT(u^2-1))$ | 2.231 | 2.375  | 1.010 |

CURRENT RATING

|                            |   |              |       |      |                               |           |           |           |
|----------------------------|---|--------------|-------|------|-------------------------------|-----------|-----------|-----------|
| Critical Temp Rise of Soil | C | IEC60287-1-1 |       | dT   | $Wk(Pc/2p)LN(Gf)$             | 0.0       | 0.0       | 0.0       |
| Top Equation               |   | IEC60287-1-1 | 1.4.1 | Etop | $DT-Wd(0.5T1+n(T2+T3+T4))-dT$ | 5.994E+01 | 5.993E+01 | 4.997E+01 |
| Bottom Equation            |   | IEC60287-1-1 | 1.4.1 | Ebot | $RT1+nR(1+l1+l2)(T3+T4)$      | 3.847E-04 | 4.463E-04 | 2.105E-04 |
| Calculated Amps            | A | IEC60287-1-1 | 1.4.1 | I    | $v(Etop/Ebot)$                | 395       | 366       | 487       |

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|--------------------|-------|--------------|--------|--------|-----------------------|----------------|-----------------|----------------|
| Calculated MVA     | MVA   | IEC60287-1-1 | 1.4.1  | MVA    | $v3UI$                | 22.6           | 20.9            | 27.8           |
| ELECTRICAL DATA    |       |              |        |        |                       |                |                 |                |
|                    |       |              |        |        |                       |                |                 |                |
| Inductance         | uH/m  | IEC60287-1-1 | 2.3    | L      | $0.2LN(2s/Dm)$        | 0.174          | 0.174           | 0.174          |
| Reactance          | uO/m  | IEC60287-1-1 | 2.3    | X      | $w Lm \times 10^{-3}$ | 54.7           | 54.7            | 54.7           |
| Pos Seq Resistance | uO/m  |              |        | R1     | $R'1(1+l1+l2)$        | 134.1          | 134.1           | 134.1          |
| Pos Seq Impedance  | uO/m  |              |        | Z1     | $v(R12+X12)$          | 144.9          | 144.9           | 144.9          |
| Capacitance        | uF/km | IEC60287-1-1 | 2.2    | C      | $e/(18LN(Di/ds))$     | 0.212          | 0.212           | 0.212          |
| THERMAL LOSS       |       |              |        |        |                       |                |                 |                |
|                    |       |              |        |        |                       |                |                 |                |

|              |             |                              |              |  |  |  |  |  |
|--------------|-------------|------------------------------|--------------|--|--|--|--|--|
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|                         | Unit  | Reference    | Clause  | Symbol | Formula                    | TREFOIL GND | TREFOIL DUCT | TREFOIL AIR |
|-------------------------|-------|--------------|---------|--------|----------------------------|-------------|--------------|-------------|
| Thermal Resistant Ins   | K.m/W | IEC60287-2-1 |         | T1     | $(p/2p)LN[1+(2t/dc)]$      | 0.451       | 0.451        | 0.451       |
| Thermal Resistant Osh   | K.m/W | IEC60287-2-1 |         | T3     | $(p/2p)LN[1+(2t/D)]$       | 0.208       | 0.130        | 0.130       |
| Thermal Resistant Ext   | K.m/W | IEC60287-2-1 | 2.2     | T4     | $(pT/2p)LN(u+SQRT(u^2-1))$ | 2.231       | 2.375        | 1.010       |
| Thermal Resistant Total | K.m/W | Mc Graw hill | Page 34 | ST     | T1+T2+T3+T4                | 2.891       | 2.957        | 1.592       |
|                         |       |              |         |        |                            |             |              |             |
| Temp Over Ins           | C     | Mc Graw hill | Page 34 | T1     | T1=W*T1                    | 9.45        | 9.42         | 14.40       |
| Temp Over Osh           | C     | Mc Graw hill | Page 34 | T3     | T3=W*T3                    | 46.68       | 49.57        | 32.21       |
| Temp Ambient            | C     | Mc Graw hill | Page 34 | Ti     | Ta=W*T4                    | 30          | 30           | 40          |
|                         |       |              |         |        |                            |             |              |             |
| Losses - Conductor      | W/m   | IEC60287-1-1 | 2.4.2   | Wc     | 3I2R                       | 19.83       | 17.06        | 30.20       |
| Losses - Dielectric     | W/m   | IEC60287-1-1 | 2.2     | Wd     | 3wCUo2tand                 | 0.024       | 0.024        | 0.024       |
| Losses - Metal sheath   | W/m   | IEC60287-1-1 |         | Wm     | Wcl1                       | 1.07        | 3.78         | 1.66        |
|                         |       |              |         |        |                            |             |              |             |